Nucleic Acids

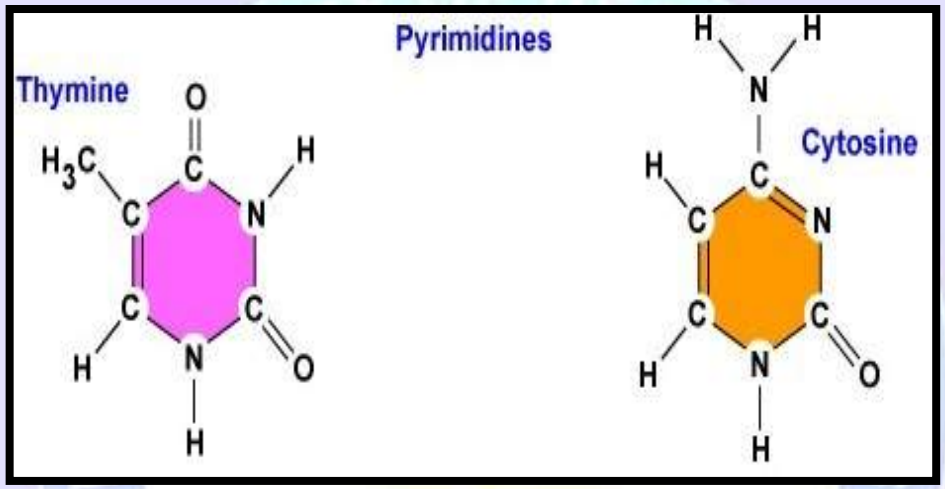
LECTURE OBJECTIVES

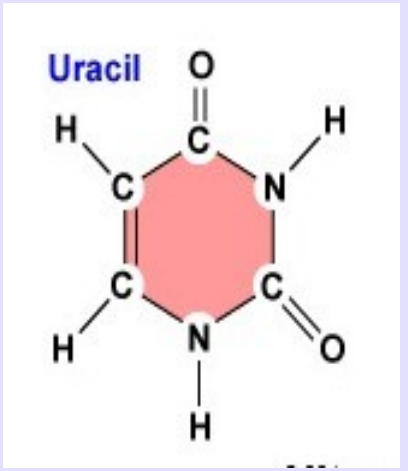
At the end of lecture the student should;

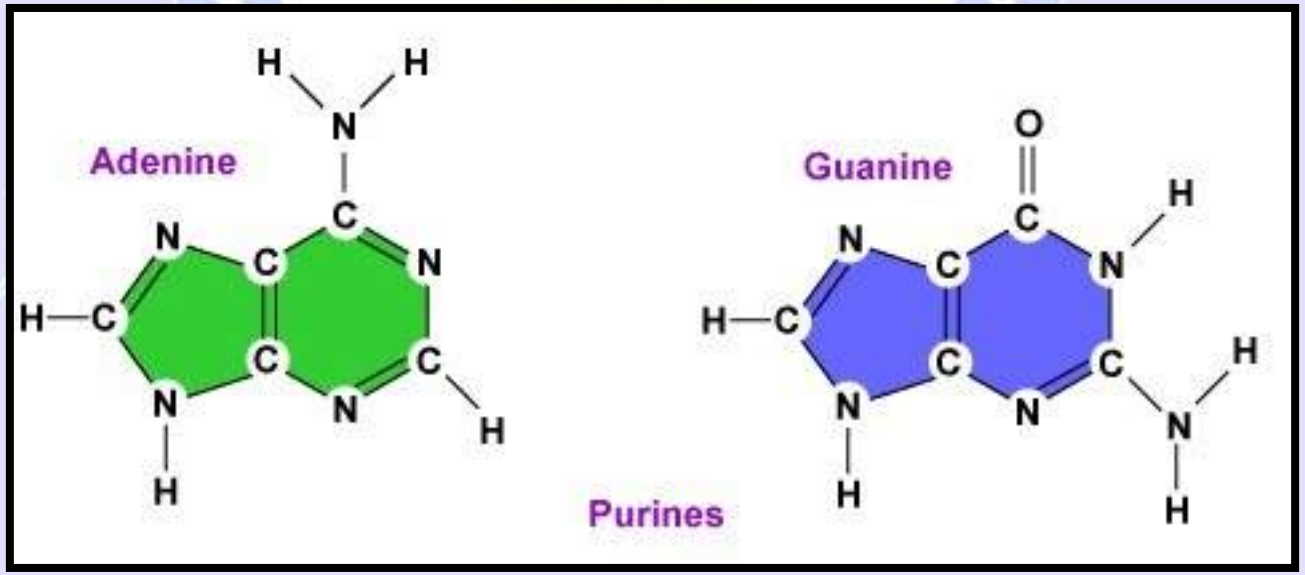
- Enlist nitrogenous bases present in the nucleic acids
- Describe the structure of nucleoside and nucleotides
- Discuss the functions of nucleotides
- Describe the structure of DNA and RNA
- Differentiate between DNA and RNA
- Discuss the packaging of DNA in chromosomes
- Explain polymerization of nucleotides via 3’-5’ phosphodiester bonds

NITROGENOUS BASES

- Cyclic compounds having nitrogen
- Non protein





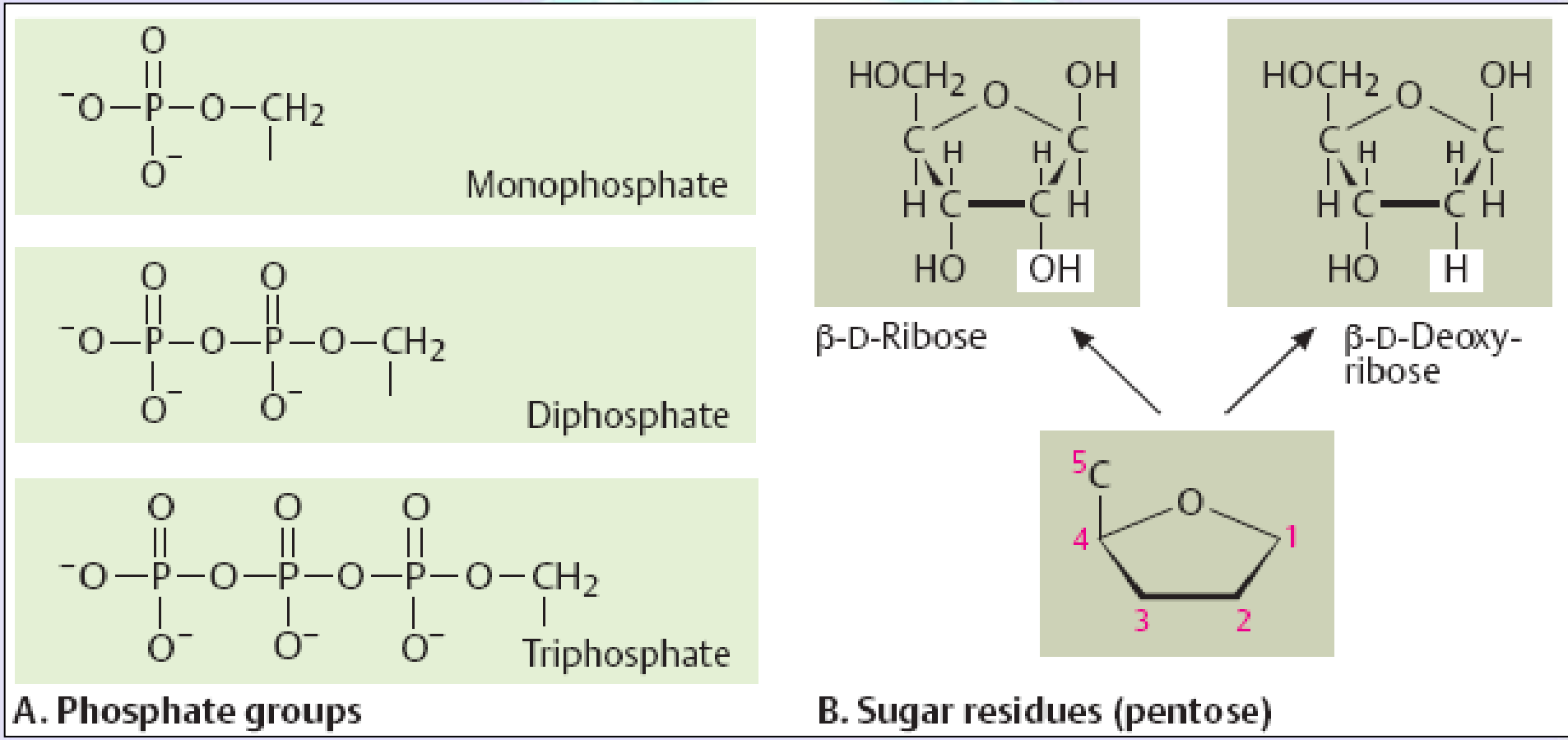


- Made form amino acids

- Purines
  - Two rings
- Pyrimidines
  - One ring

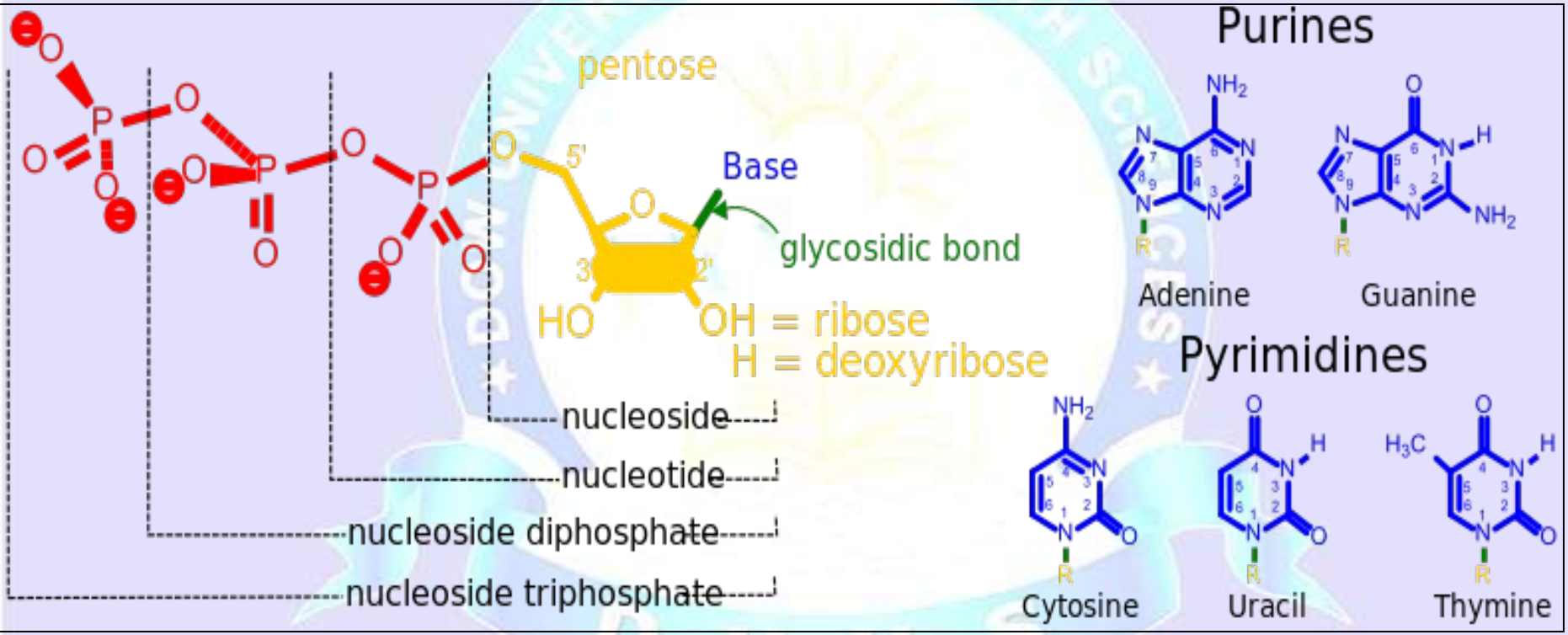
Pyrimidine CUT (cytosine, uracil & thymine) from purines.

SUGAR AND PHOSPHATE



Name the sugar residue with a prime (‘) mark

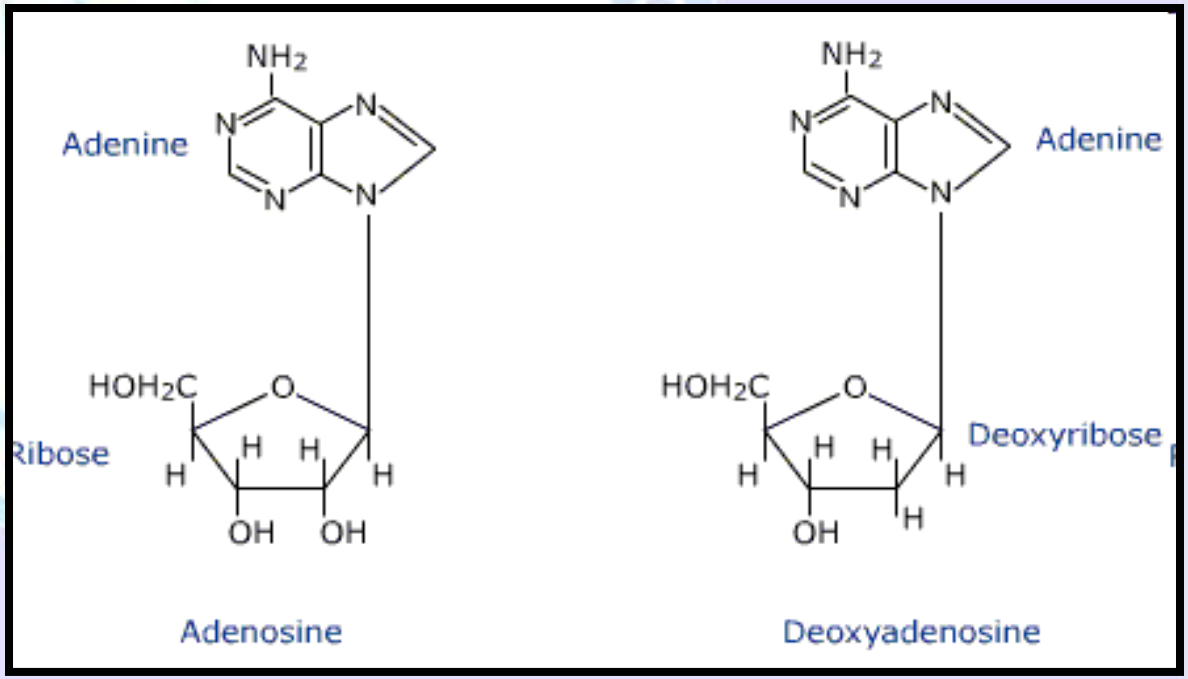
NUCLEOSIDE AND NUCLEOTIDE



- A **nucleoside** consists simply of a nucleobase (also termed a nitrogenous base) and a five-carbon sugar (either ribose or deoxyribose), whereas a **nucleotide** is composed of a nucleobase, a five-carbon sugar, and one or more phosphate groups.

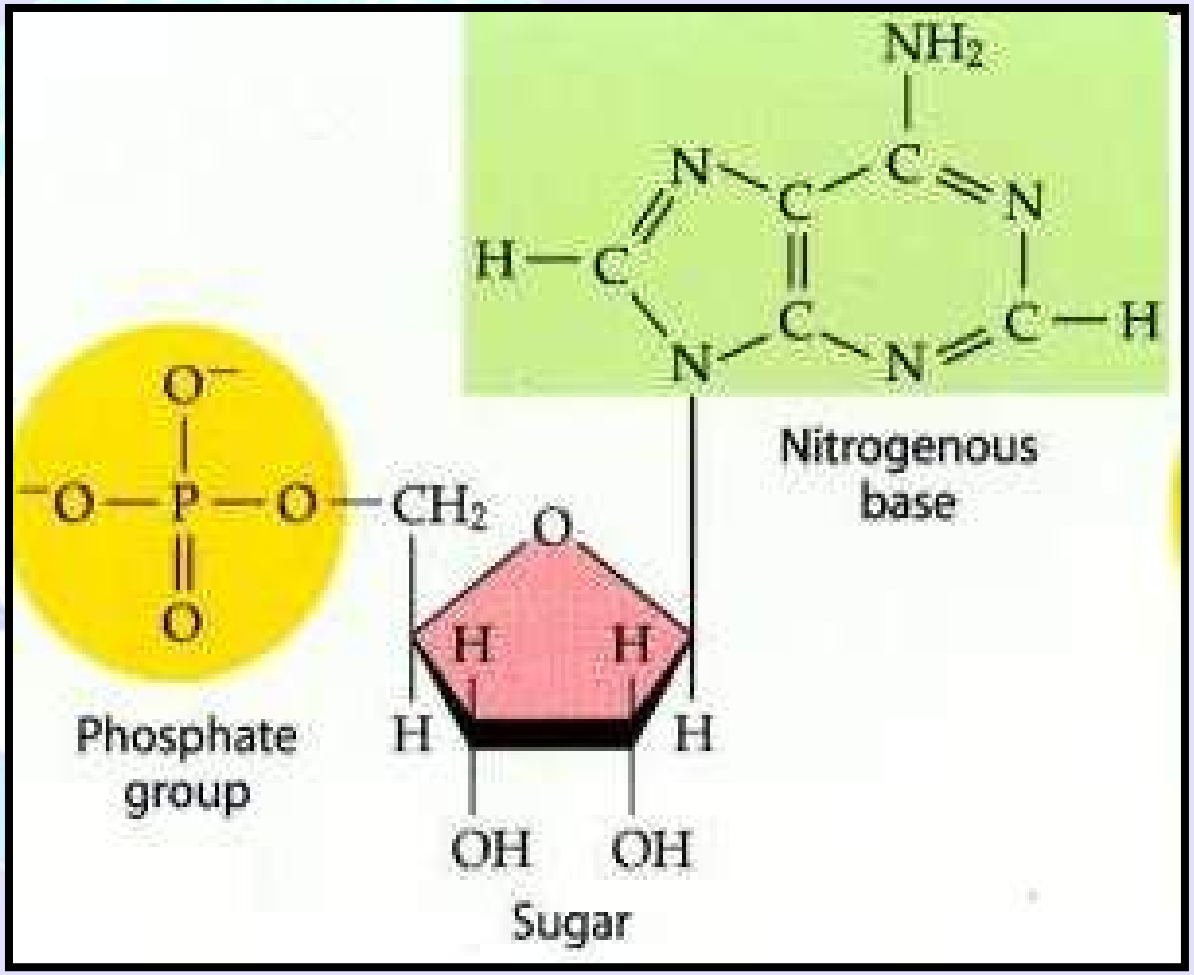
STRUCTURES OF NUCLEOSIDES

- Sugar
  - Ribose
  - Deoxyribose
- Base
  - Purine
  - Pyrimidine
- Name
  - Adininine→Adinosine
  - Guanine →Guanosine
  - Thymine →Thymidine
  - Uracil →Uridine
  - Cytosine →Cytidine



STRUCTURES OF NUCLEOTIDES

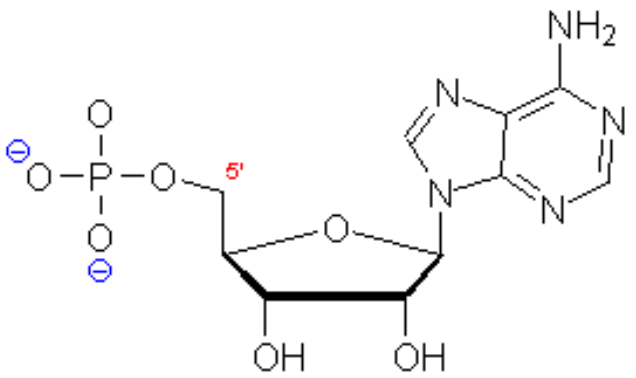
- Sugar
  - Ribose
  - Deoxyribose
- Base
  - Purine
  - Pyrimidine
- Phosphate
  - One-mono
  - Two-Di
  - Three-Tri



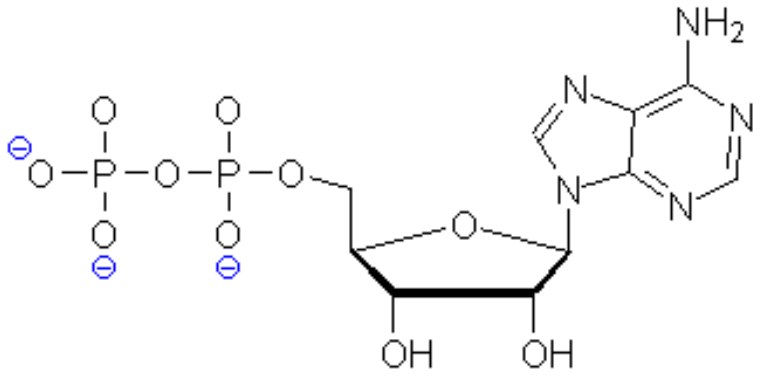
FUNCTIONS OF NUCLEOTIDES

- Energy carriers
- Electron carriers
- Coenzymes
- Activation of metabolites
- Expression of genetic information
- Transmission of genetic information

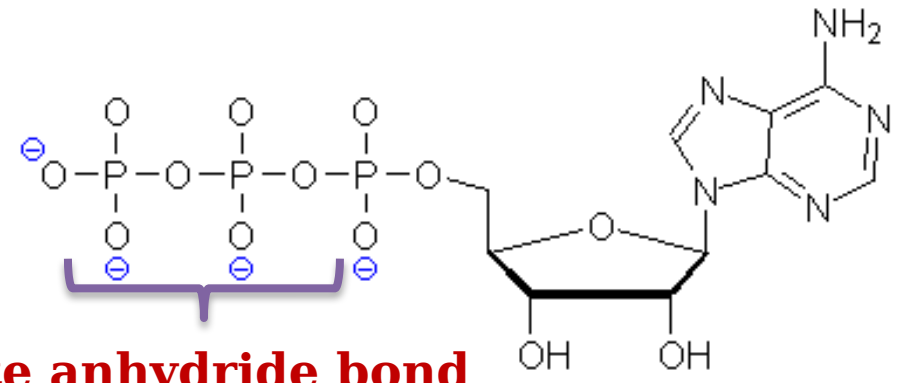
Nucleotide: Energy Carrier



Adenosine 5'-monophosphate
(AMP)



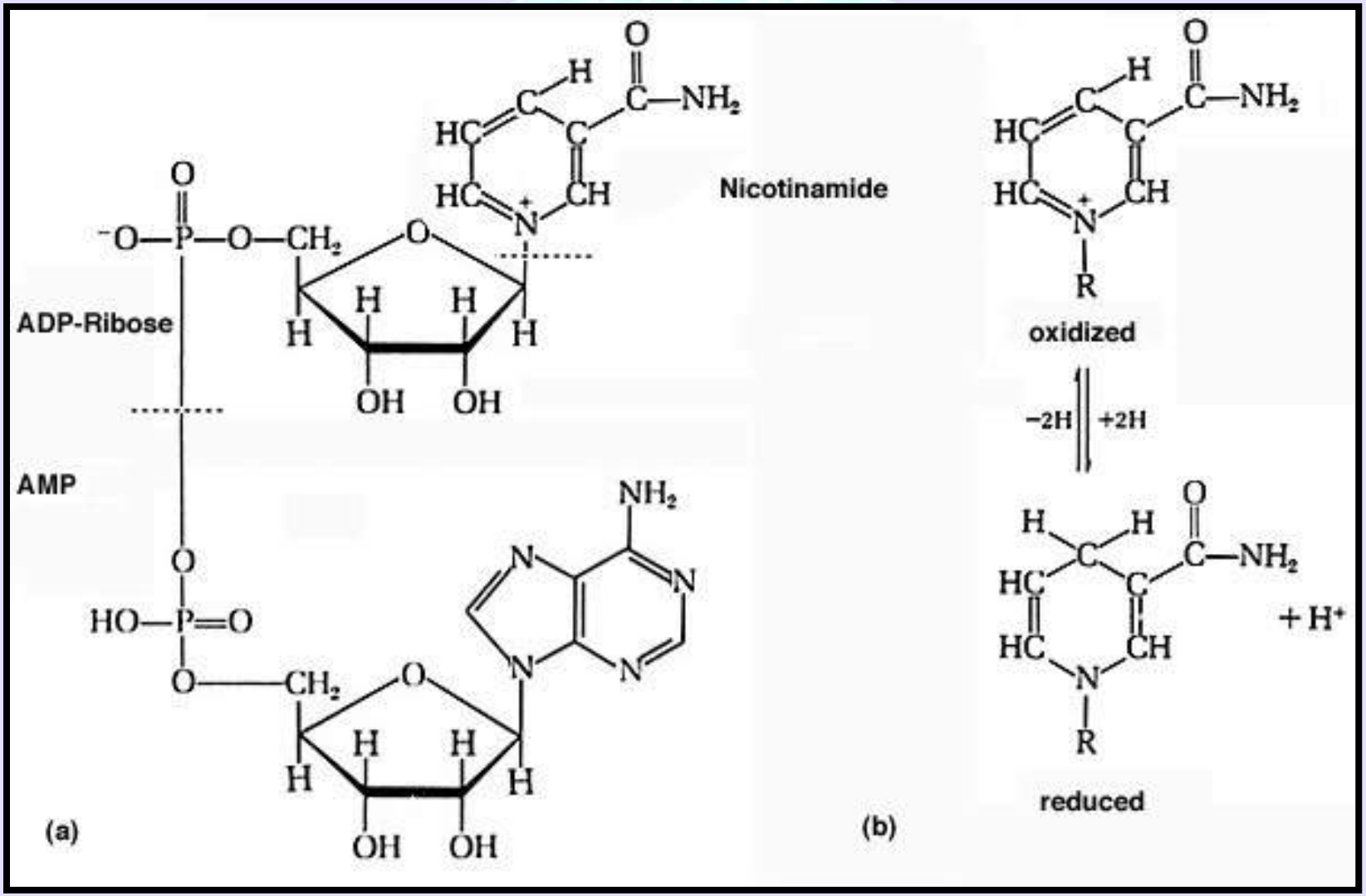
Adenosine 5'-diphosphate
(ADP)



Adenosine 5'-triphosphate
(ATP)

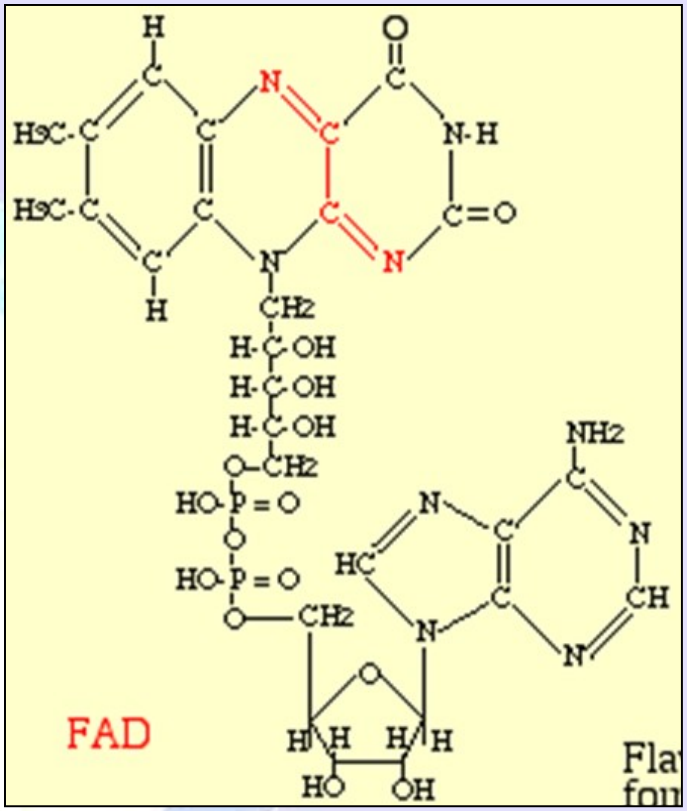
Phosphate anhydride bond
High energy
~7.4Kcal/mol

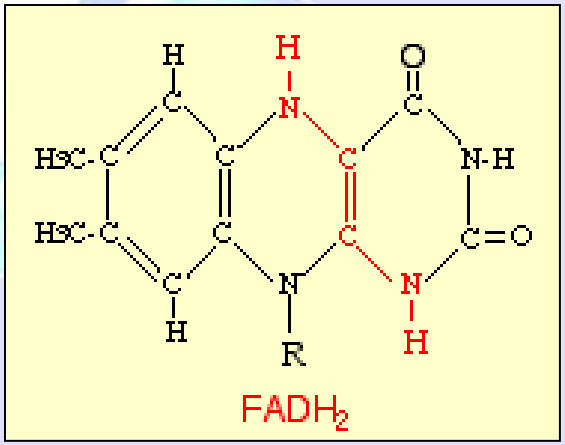
ELECTRON CARRIERS

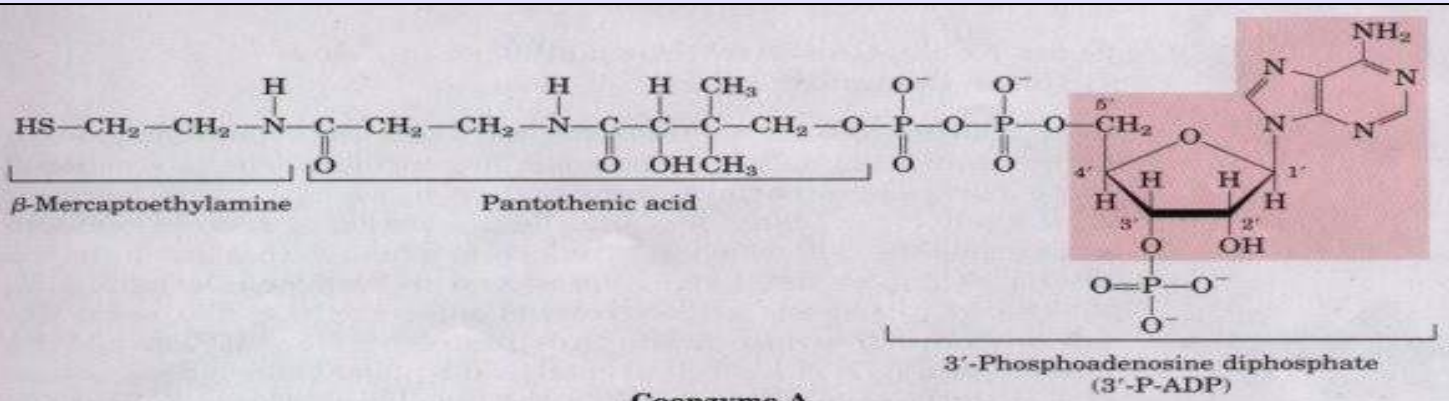


CO-ENZYMES

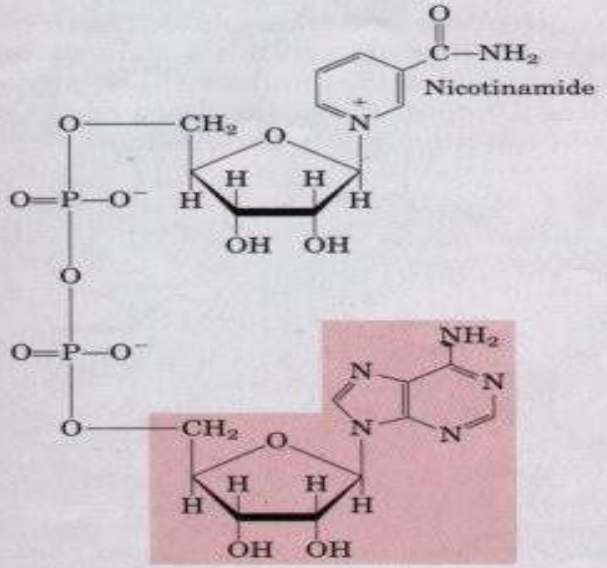
- Assist the enzymes
  - Pyruvate dehydrogenase
- Electron acceptors
- NAD, NADP
- FAD
- FMN



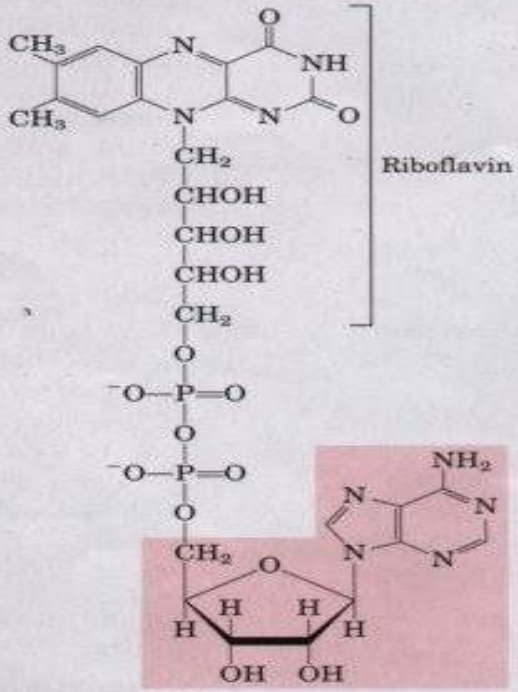




Coenzyme A

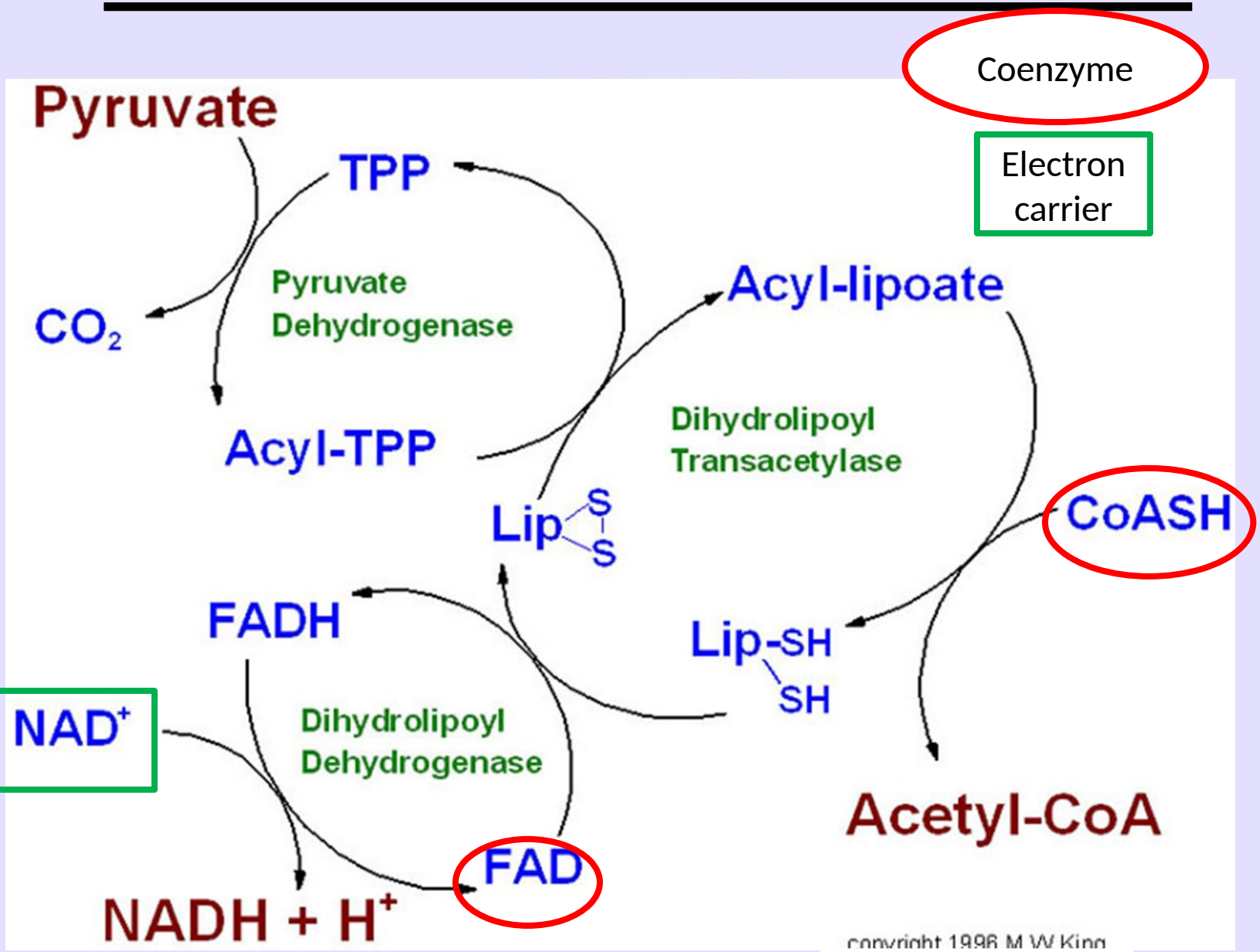


Nicotinamide adenine dinucleotide (NAD^+)



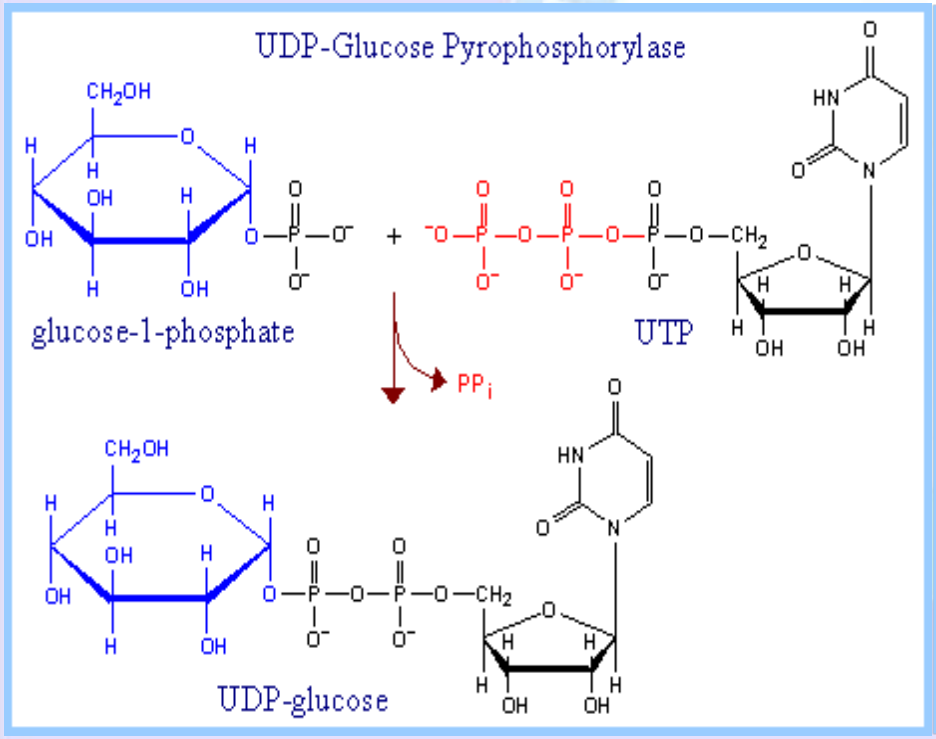
Flavin adenine dinucleotide (FAD)

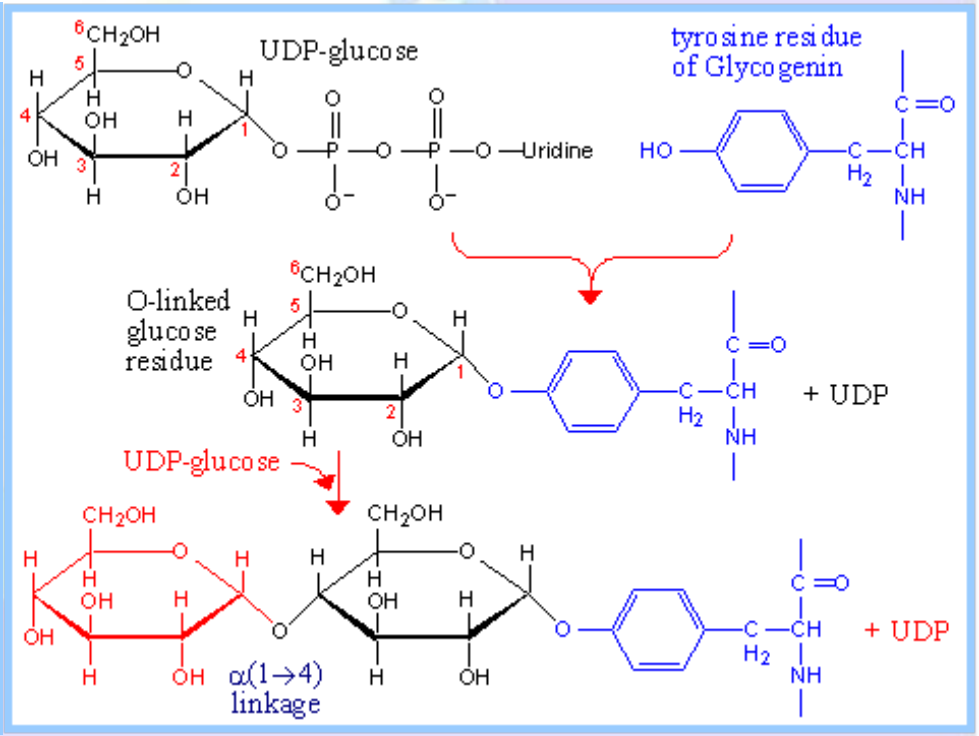
PYRUVATE DEHYDROGENASE



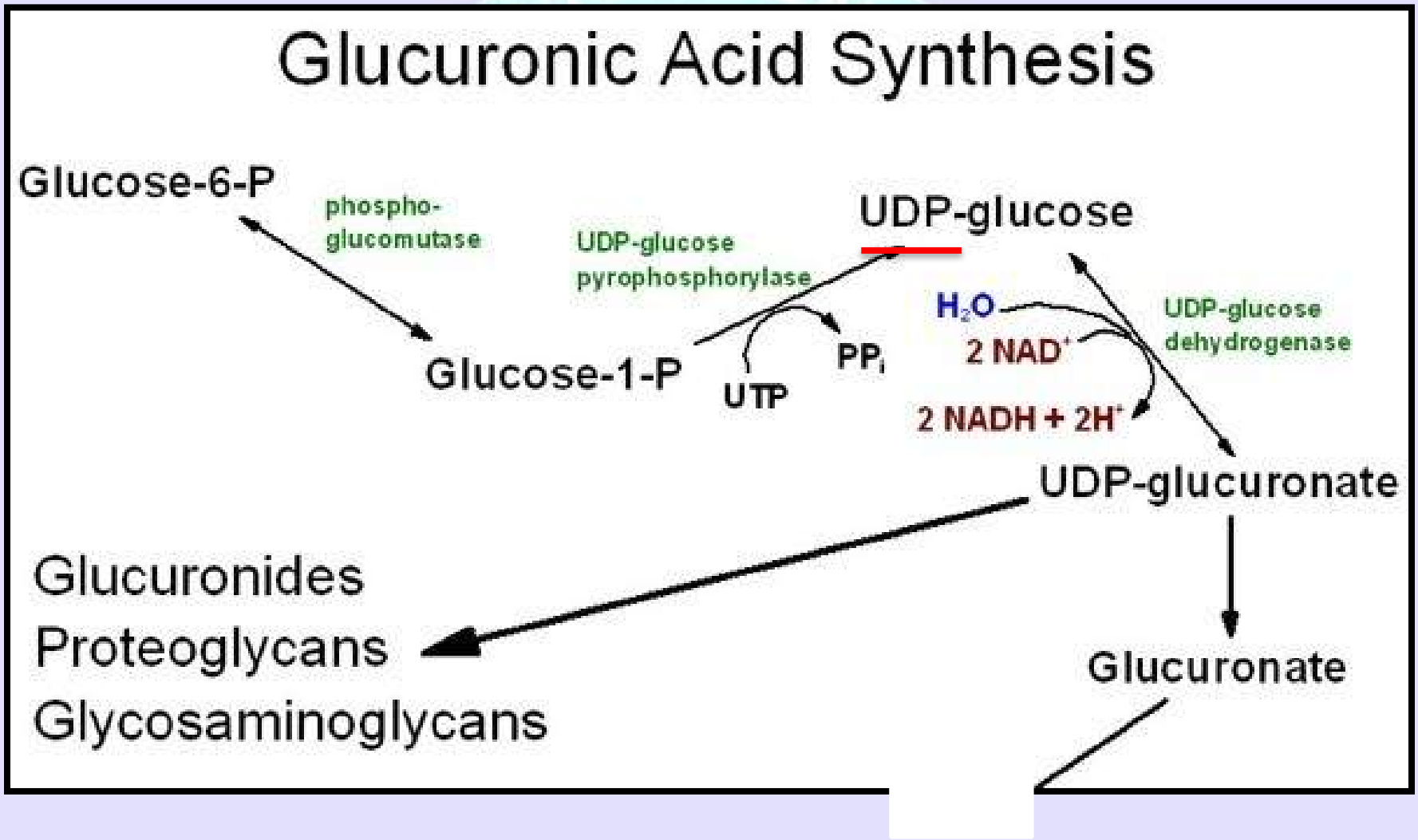
ACTIVATION OF METABOLITES

Glucose 1 Phosphate → UDP glucose → Glycogen



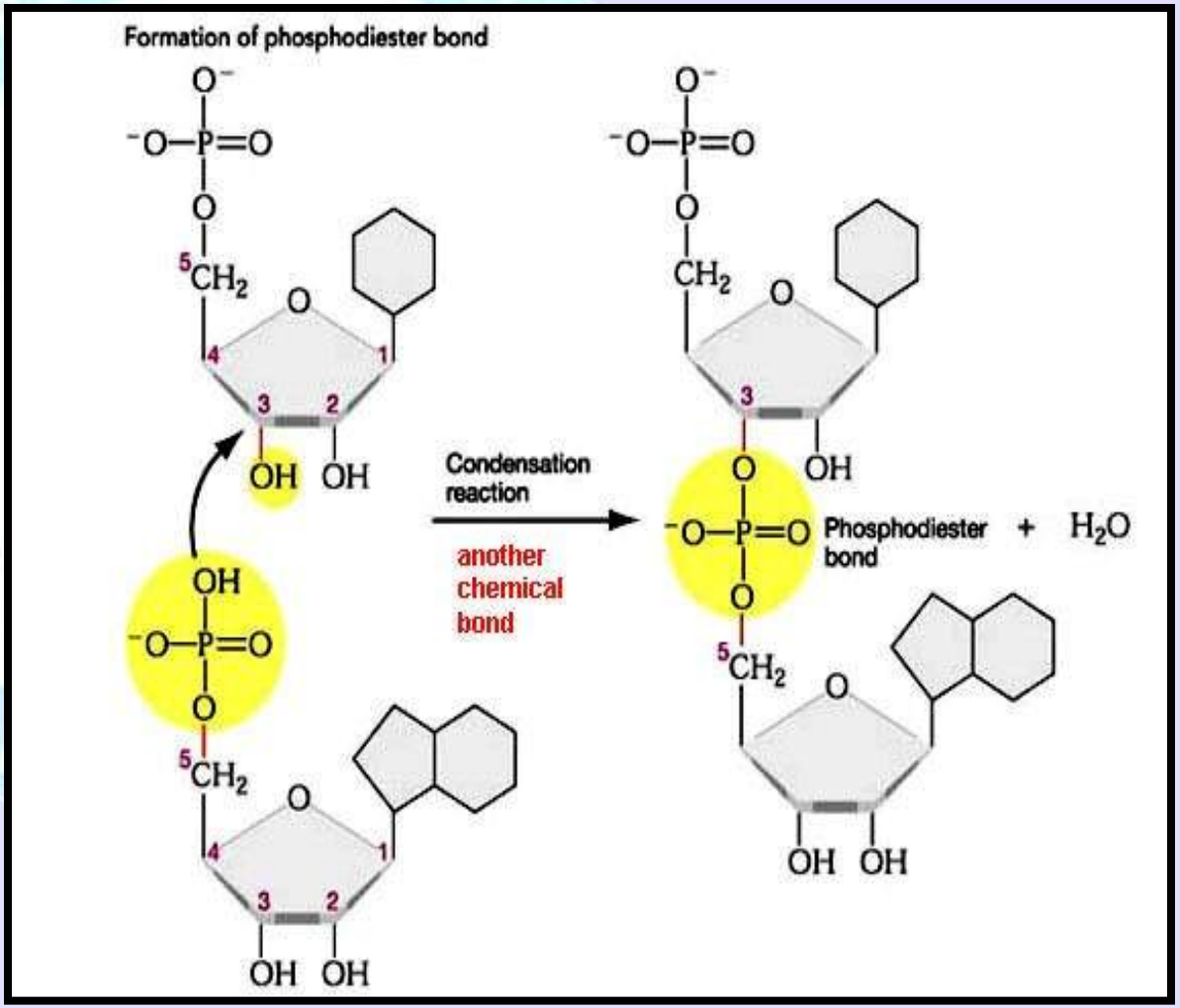


ACTIVATION OF METABOLITES



NUCLEIC ACIDS

- Nucleotides polymerize
- Sugar phosphate backbone
- Phosphodiester bond

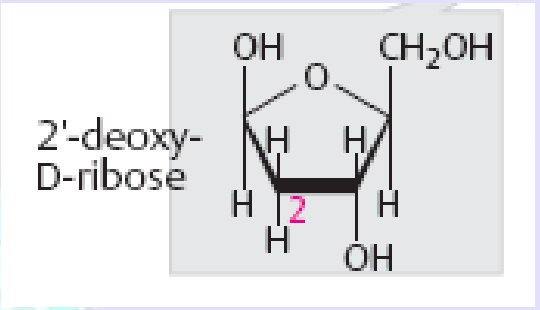


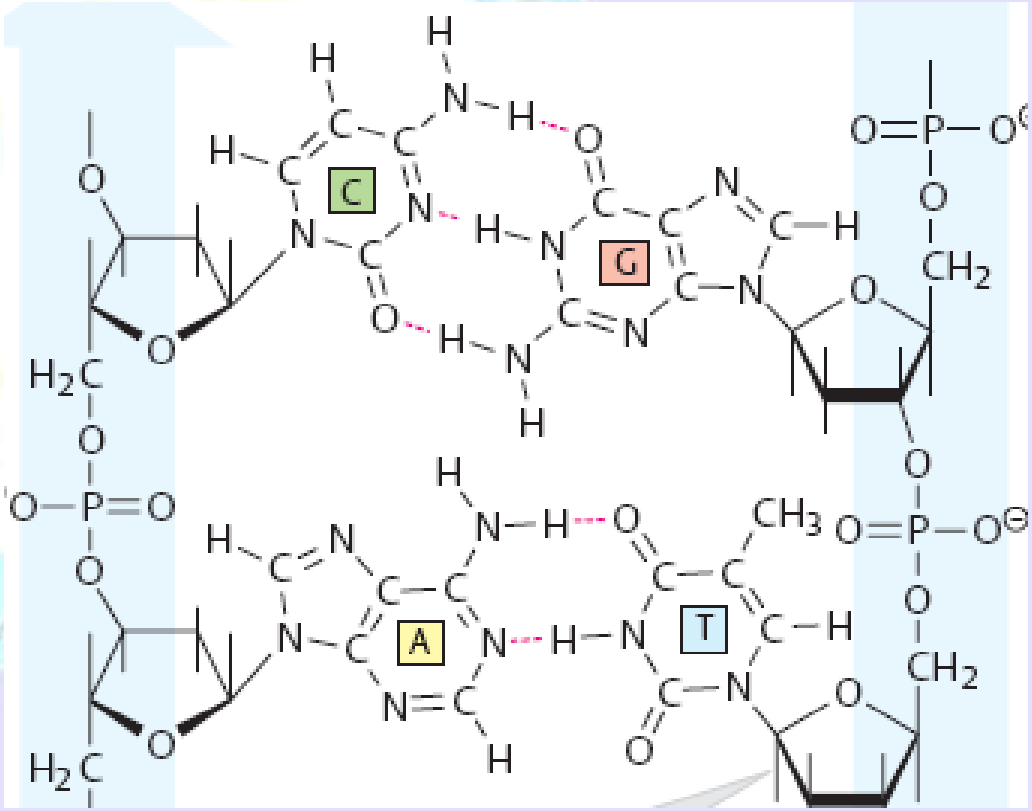
EXPRESSION OF GENETIC INFORMATION: NUCLEIC ACIDS

- Nucleotides polymerize
- 3’-5’phosphodiester bonds
- Sugar phosphate back bone
- Bases project inward
- DNA and RNA
- Nitrogenous bases differ

Deoxyribonucleic Acid

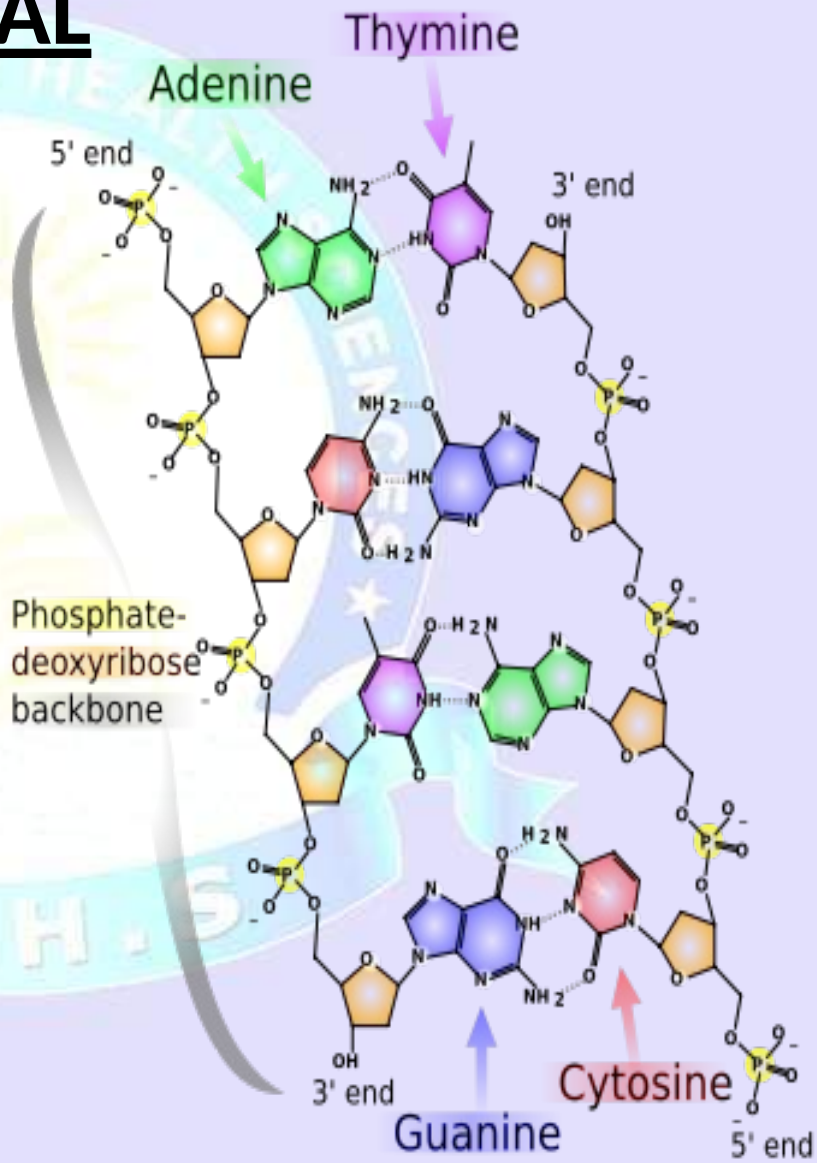
- Polynucleotide
- Deoxyribose
- Bases
  - Purine
    - adenine
    - guanine
  - Pyrimidine
    - cytosine
    - thymine
- Mono phosphate
- Double stranded





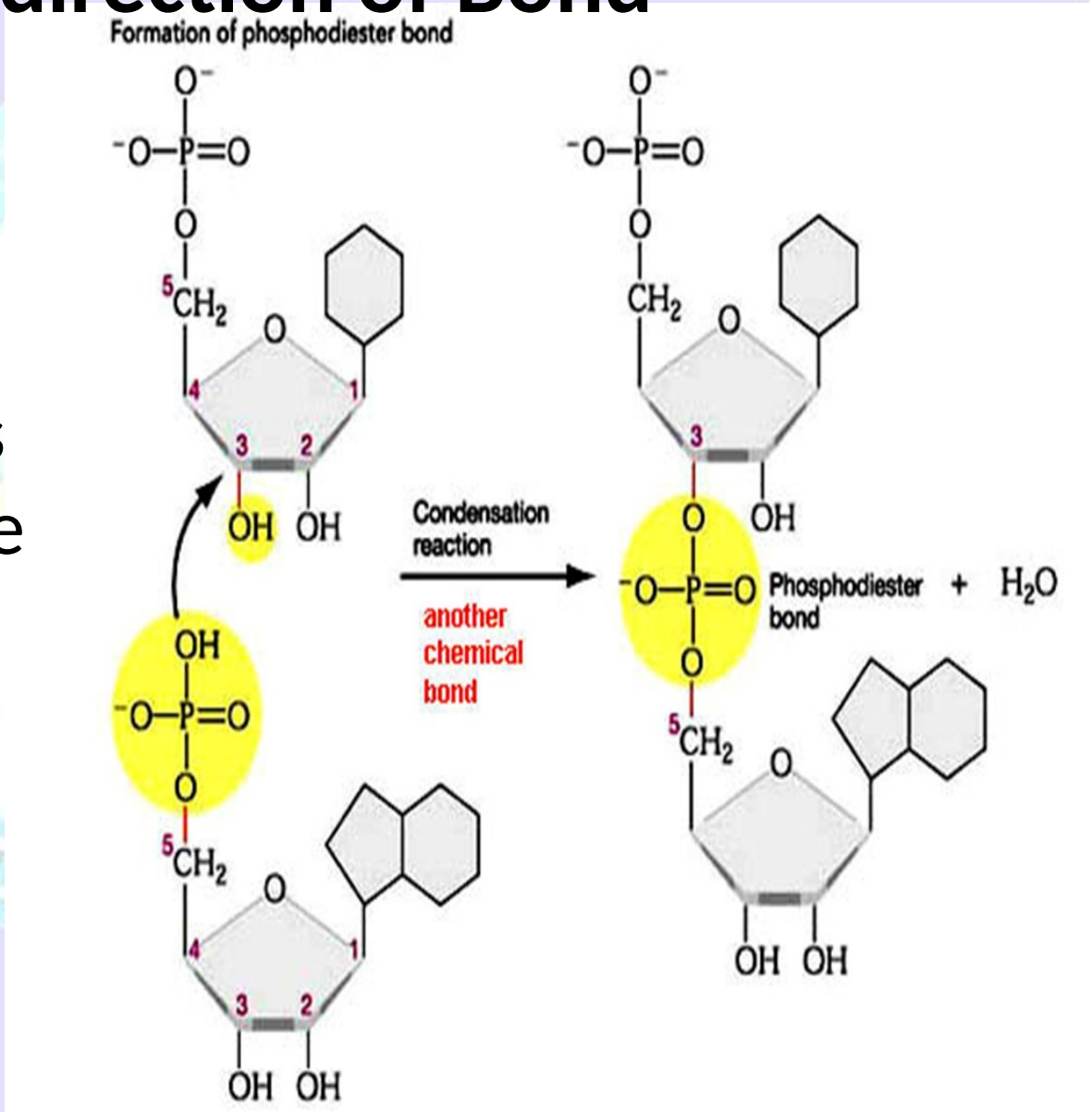
ORGANIZATION OF GENETIC MATERIAL

- Sugar Phosphate Backbone
  - Hydrophilic
- Phosphodiester Bond
- Bases Project Inward
  - Hydrophobic



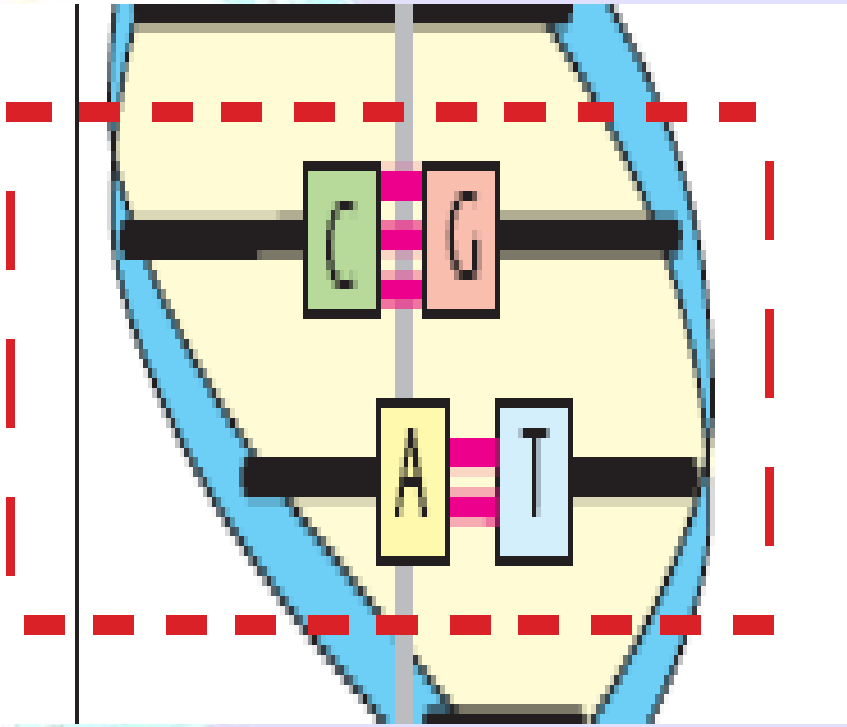
Formation and direction of Bond

- 3’ – 5’ phosphodiester bond
- Start with always free 5’ phosphate group
- End with always free 3’ hydroxyl group



Structure of Deoxyribonucleic Acid

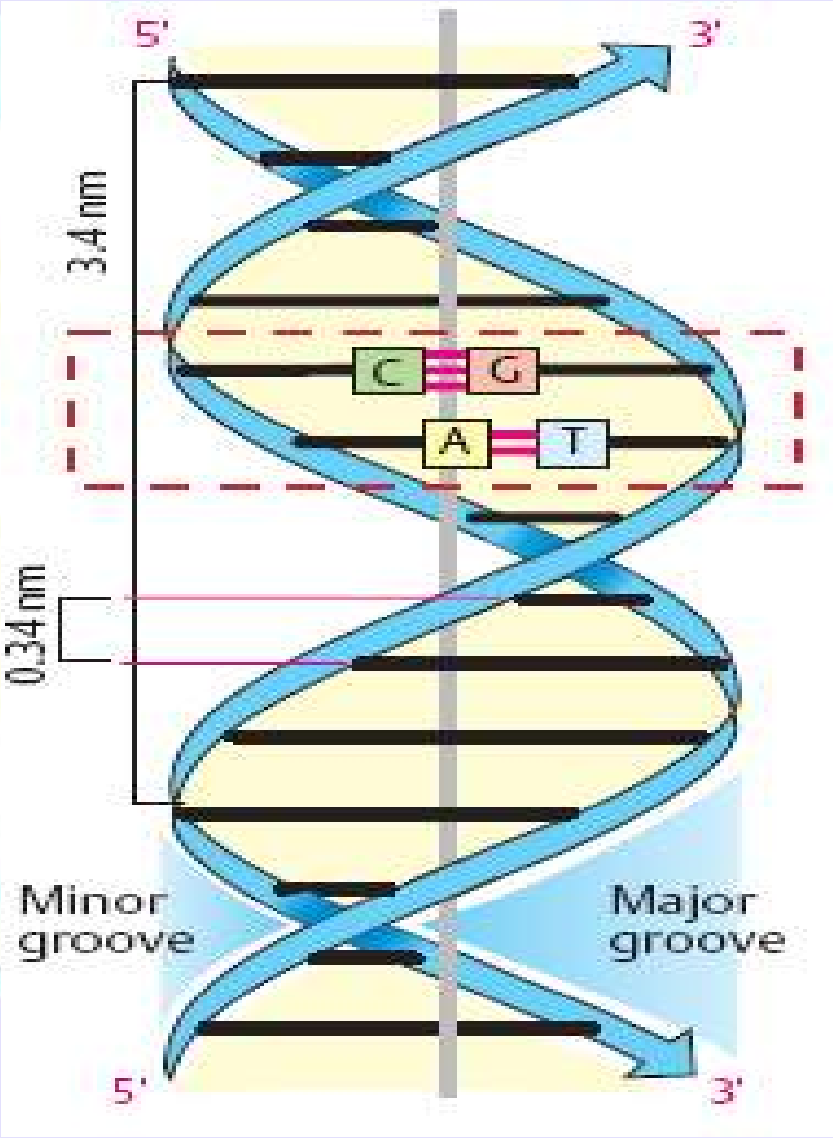
- **Base pairing**
  - **Purine → Pyrimidine**
    - **Adenine → Thymine**
    - **Guanine → Cytosine**



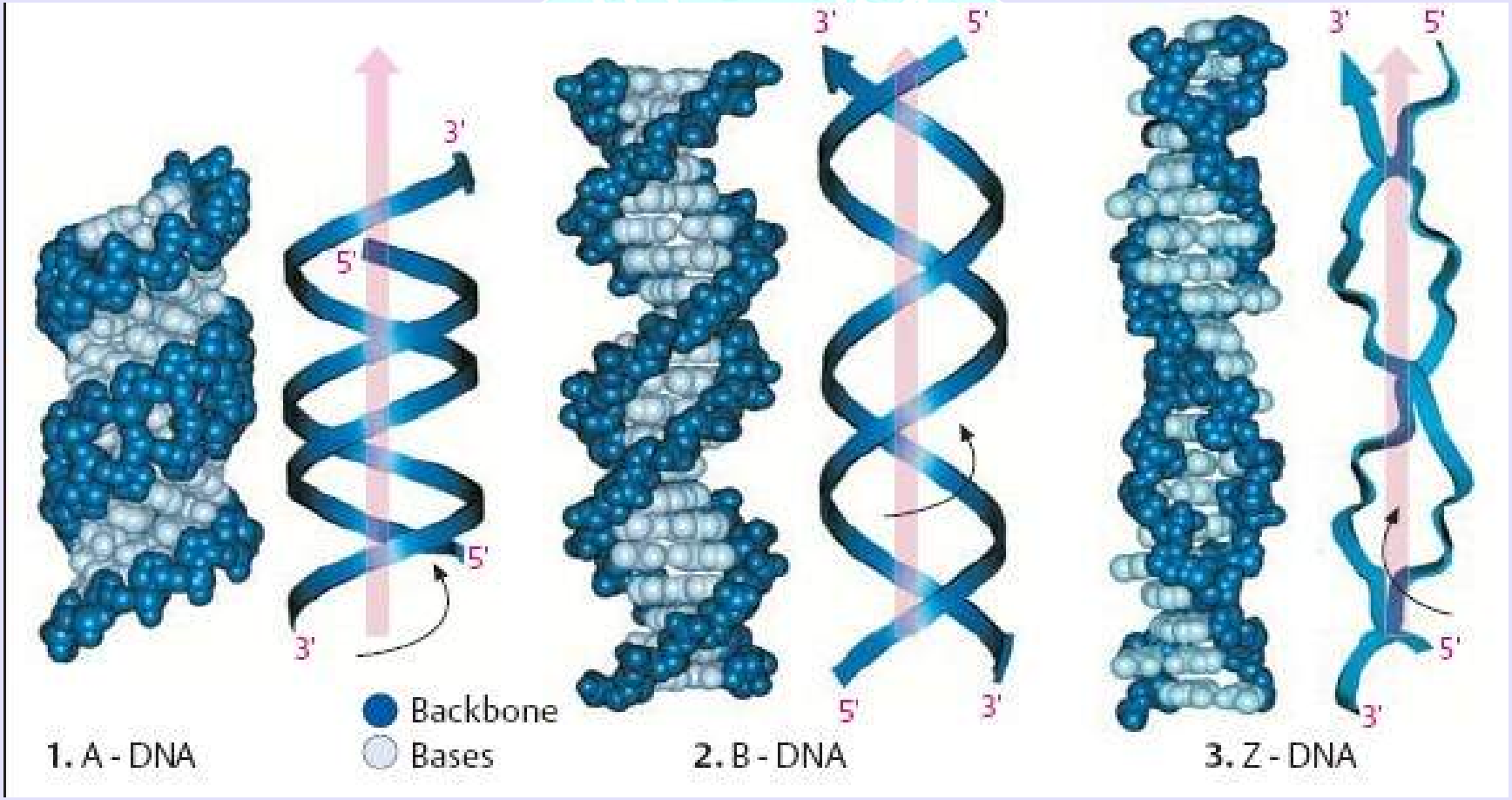
- **Base stabilization**
  - Hydrogen bond
    - A = T
    - G = C

Structure of Deoxyribonucleic Acid

- Double stranded helix
- Central Axis
- Anti-parallel
- Major groove
- Minor groove

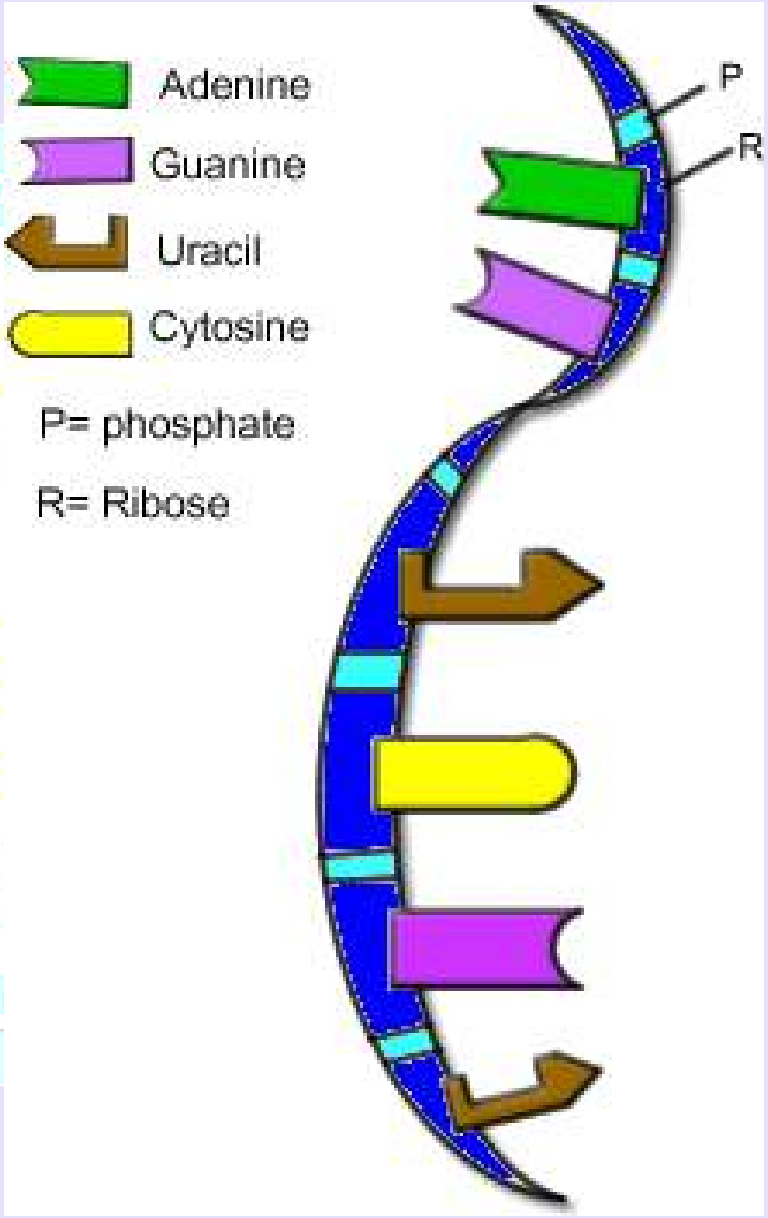


Types of DNA



Ribonucleic Acid

- Polynucleotide
- Ribose
- Bases
  - Purine
    - adenine
    - guanine
  - Pyrimidine
    - cytosine
    - uracil
- Mono phosphate
- Single stranded

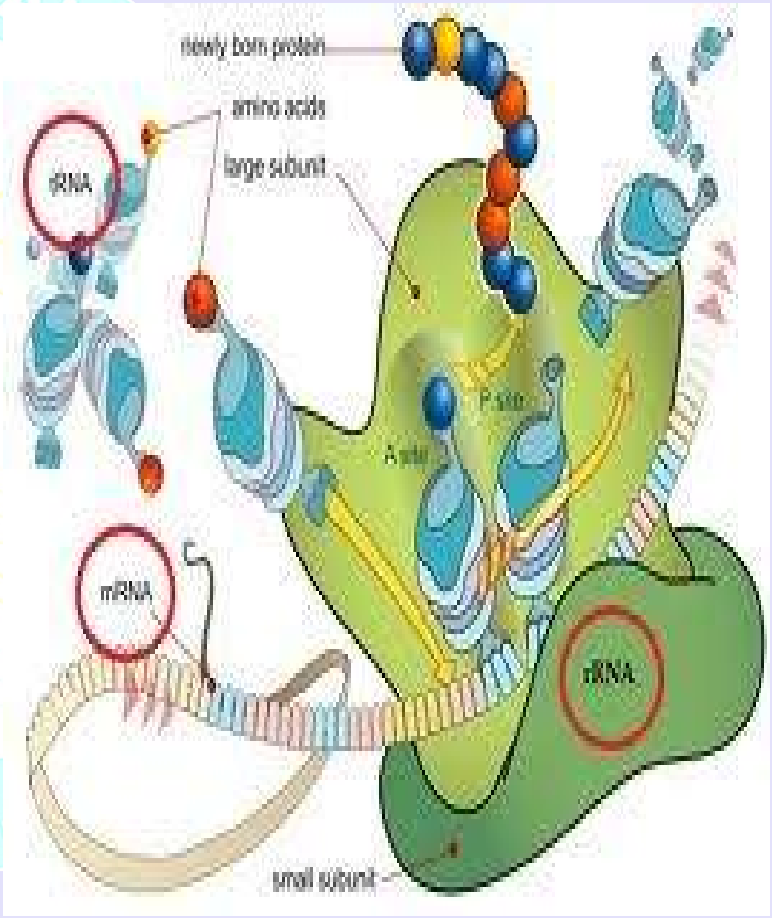


Simple Facts About DNA and Genes

- The information for development and specific function is stored in genes.
- A gene is portion of genetic information definable according to the structure and functions.
- Genes lie on chromosomes in the nuclei of the cells.
- Chromosomes are made up of long chains of DNA and proteins.

Types of RNA

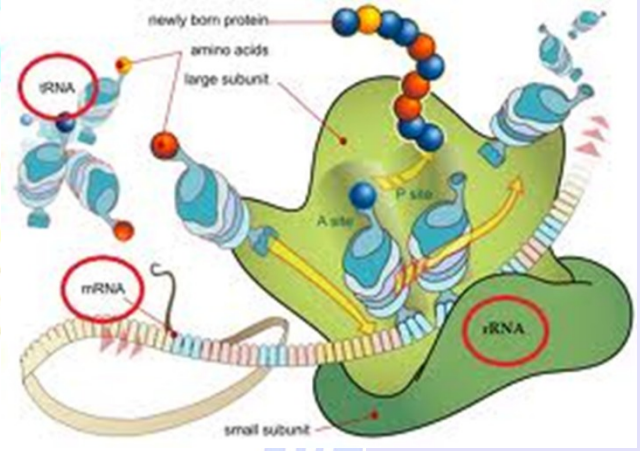
- Three major classes:
  - **mRNA (messenger)**
  - **tRNA (transfer RNA)**
  - **rRNA (ribosomal RNA)**

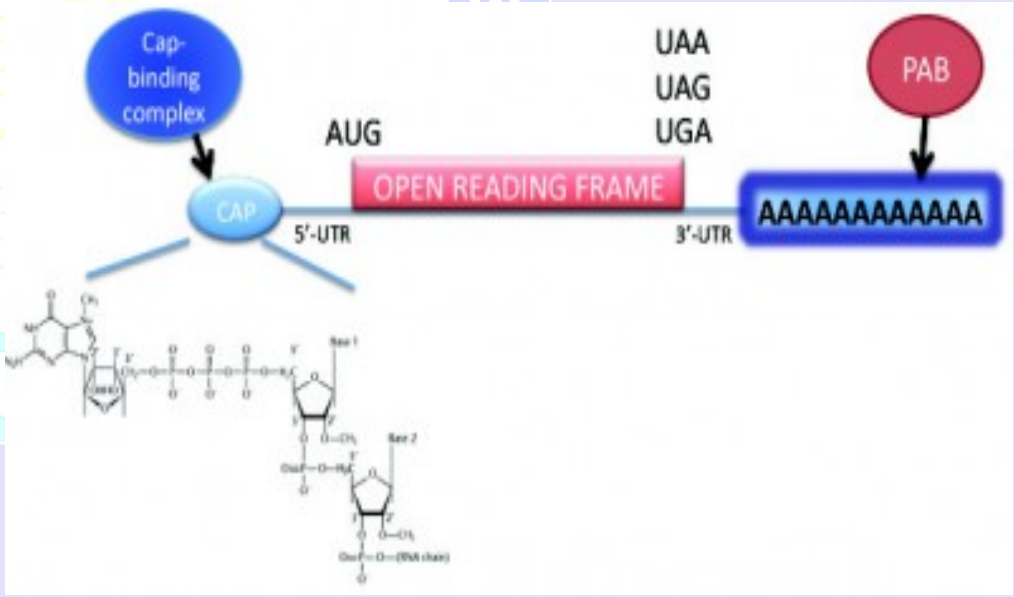


- All are synthesized from

Messenger RNA

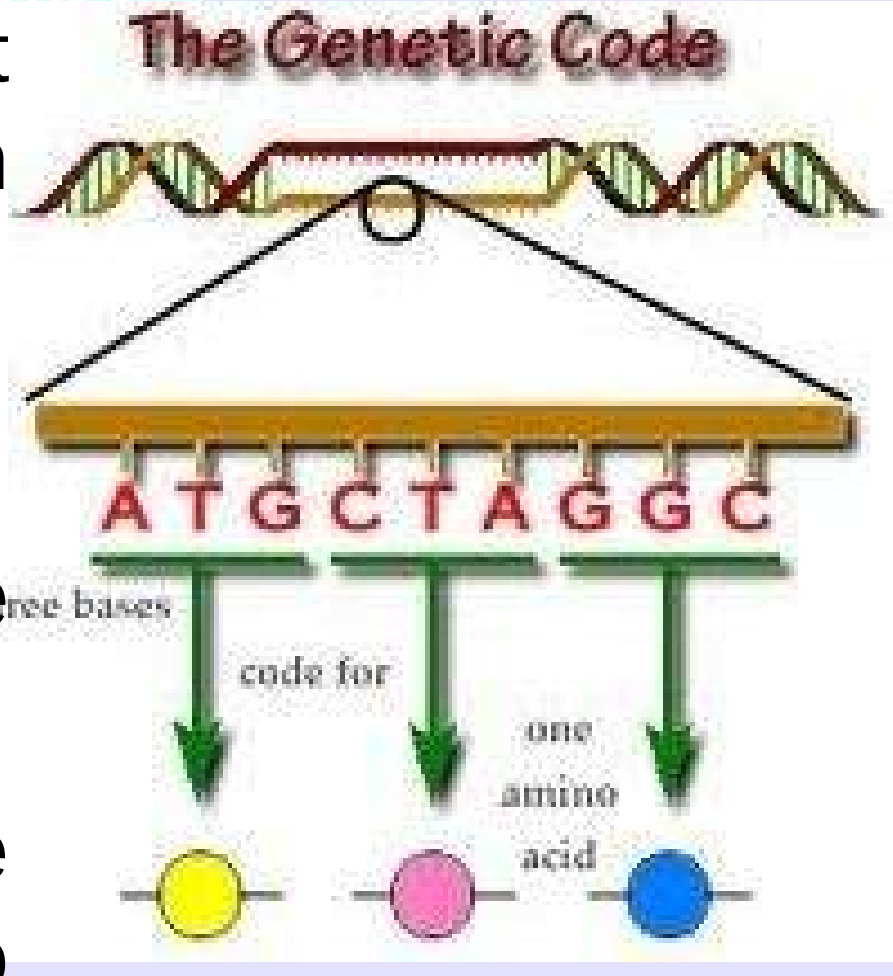
- 500 – 6000 nucleotides
- Poly A tail is present at 3’ end and a cap at 5’ end with a molecule of 7-methylguanosine
- Synthesized from DNA
- Genetic code translated
- Nucleus to cytoplasm
- 5% of total RNA.



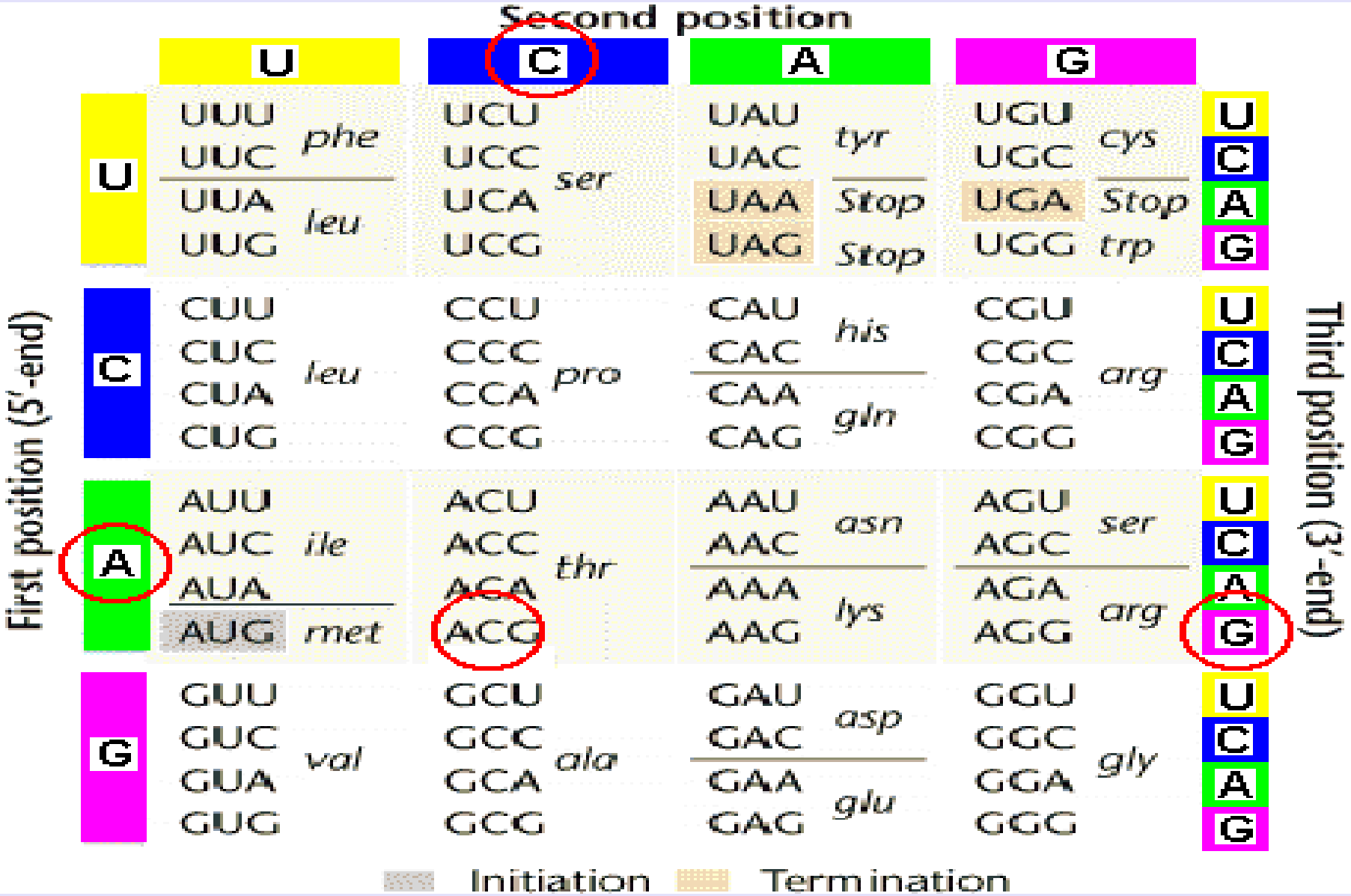


Genetic Code

- Collection of base sequence that corresponds to each amino acid.
- Triplet of nucleotide.
- Most amino acids have more than one codon.
- There are 64 possible codon codes for amino acids.



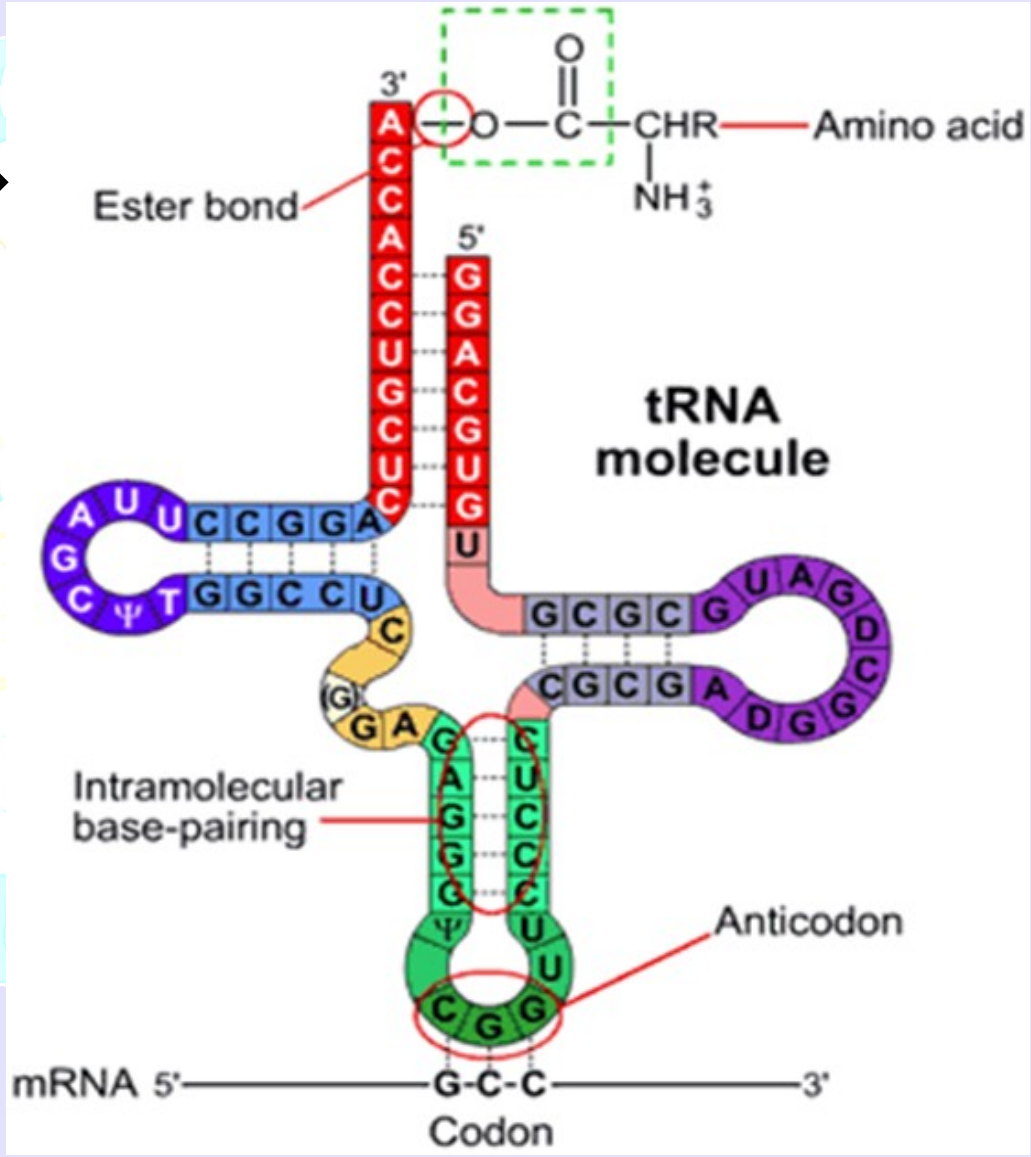
Genetic Code



Transfer RNA

- Smallest of three
- tRNA reads the code → anticodon.

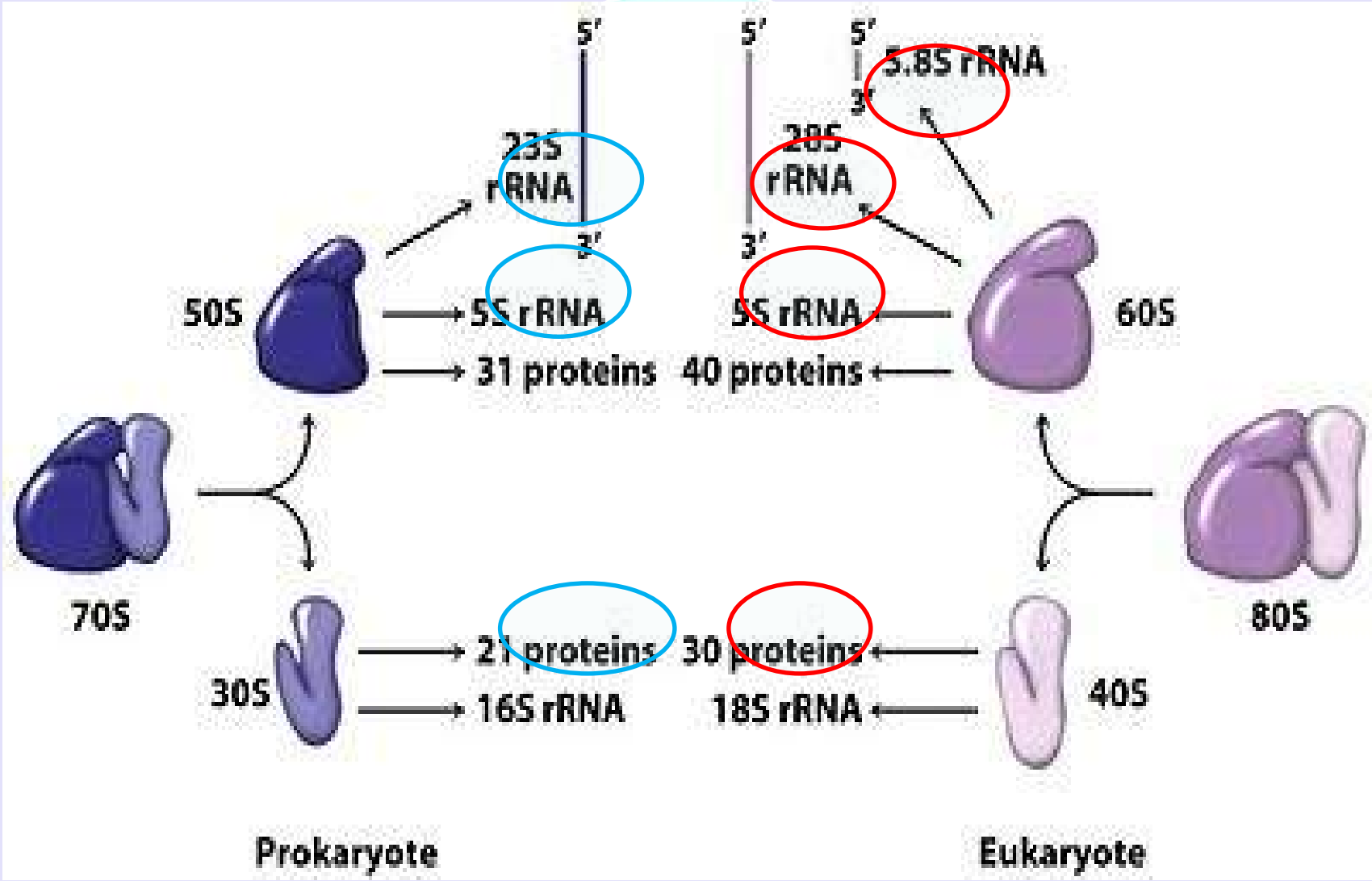
- Specfic type of tRNA for each amino acid.
- Unusual bases → pseudouracil
- Extensive intrachain base-pairing



Ribosomal RNA

- Cytoplasm.
- Ribosomal RNA + proteins → Ribosome (nucleoprotein)
- 3 distinct sizes of rRNA →In prokaryotes → 23S,16S, and 5S.
- In eukaryotes 4 sizes → 28S,18S,5.8S and 5S.
- Total 80% RNA

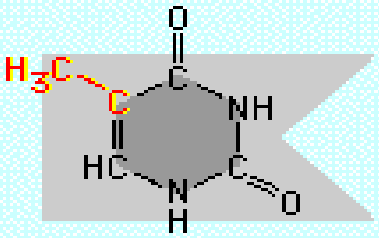
Ribosomal RNA



Differences Between RNA and DNA

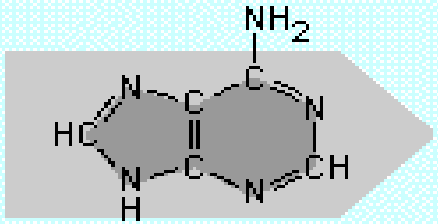
DNA

Pyrimidines

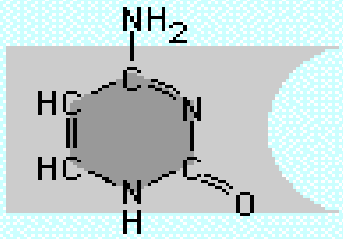


Thymine (T)

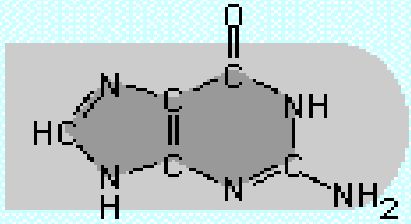
Purines



Adenine (A)



Cytosine (C)

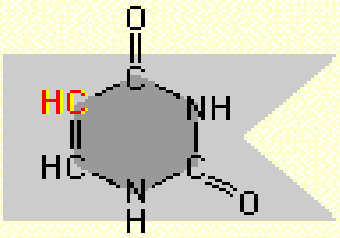


Guanine (G)

DNA bases

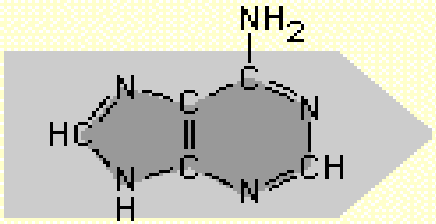
RNA

Pyrimidines

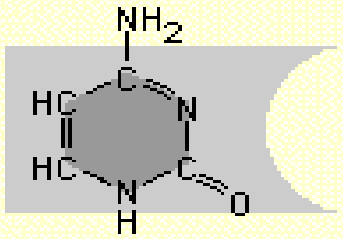


Uracil (U)

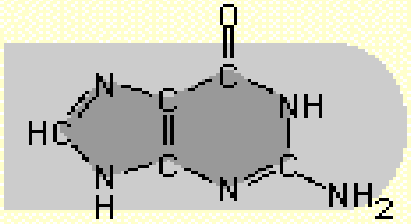
Purines



Adenine (A)

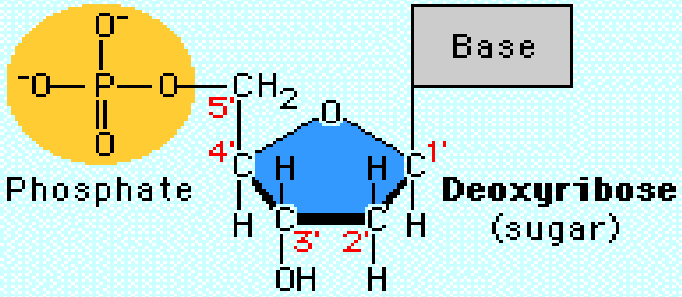


Cytosine (C)

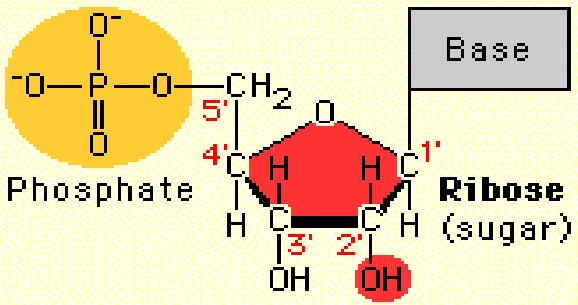


Guanine (G)

RNA bases

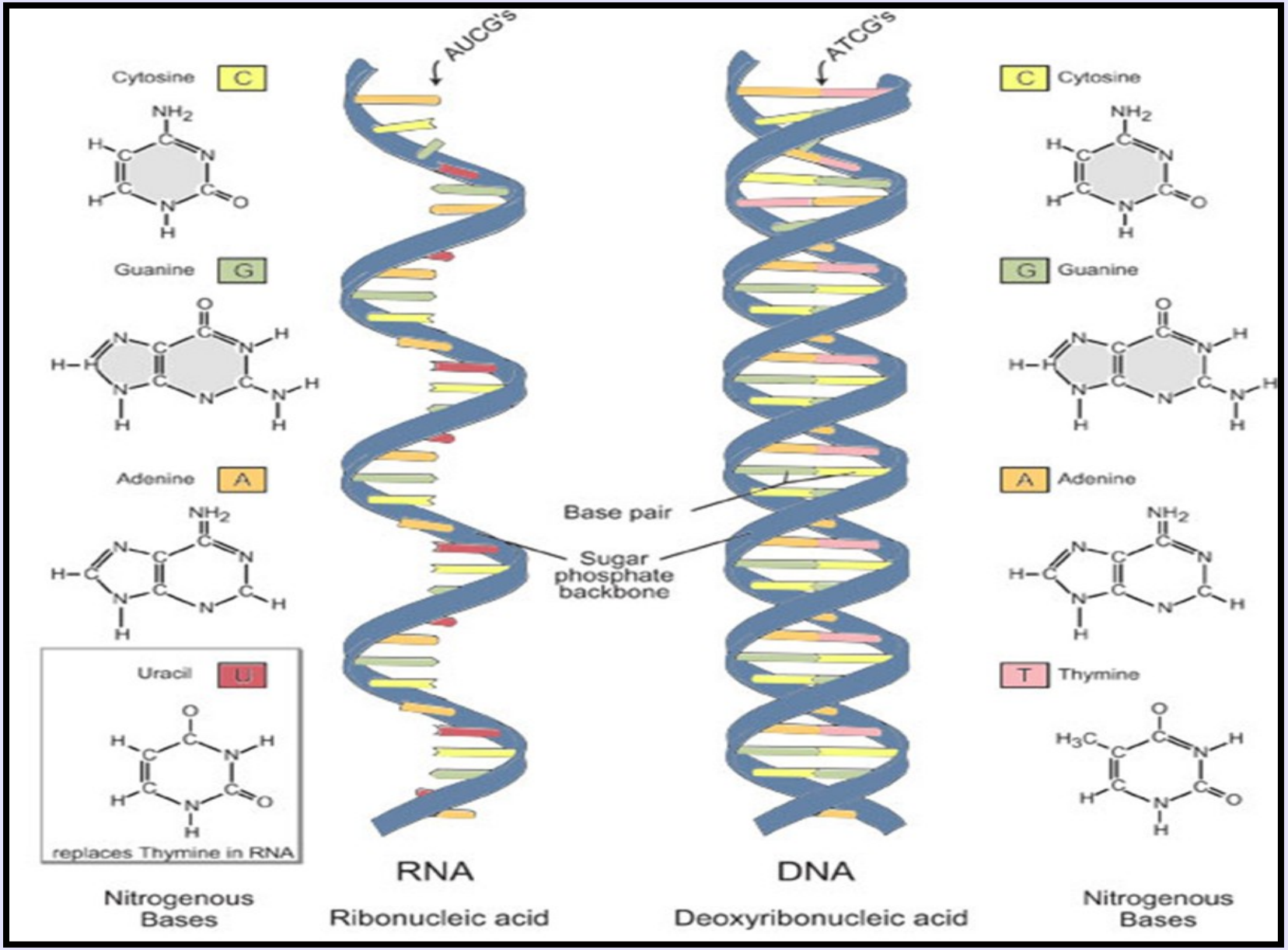


DNA nucleotide



RNA nucleotide

DNA AND RNA


AUCG's
ATCG's
Cytosine C
Guanine G
Adenine A
Uracil U
replaces Thymine in RNA
Base pair
Sugar phosphate backbone
RNA
Ribonucleic acid
DNA
Deoxyribonucleic acid
Nitrogenous Bases
C Cytosine
G Guanine
A Adenine
T Thymine
Nitrogenous Bases

Differences in DNA and RNA

DNA	RNA
• Double stranded • Sugar: deoxyribose • Pyrimidine: thymine • Location: Nucleus and Mitochondria • Synthesis; replication • Nucleoprotein; Chromosome	• Single stranded • Sugar: ribose • Pyrimidine: uracil • Location: Nucleus, Mitochondria and Cytoplasm • Synthesis; transcription • Nucleoprotein;

References

- Harper's Illustrated Biochemistry, 28th Edition, McGraw Lange

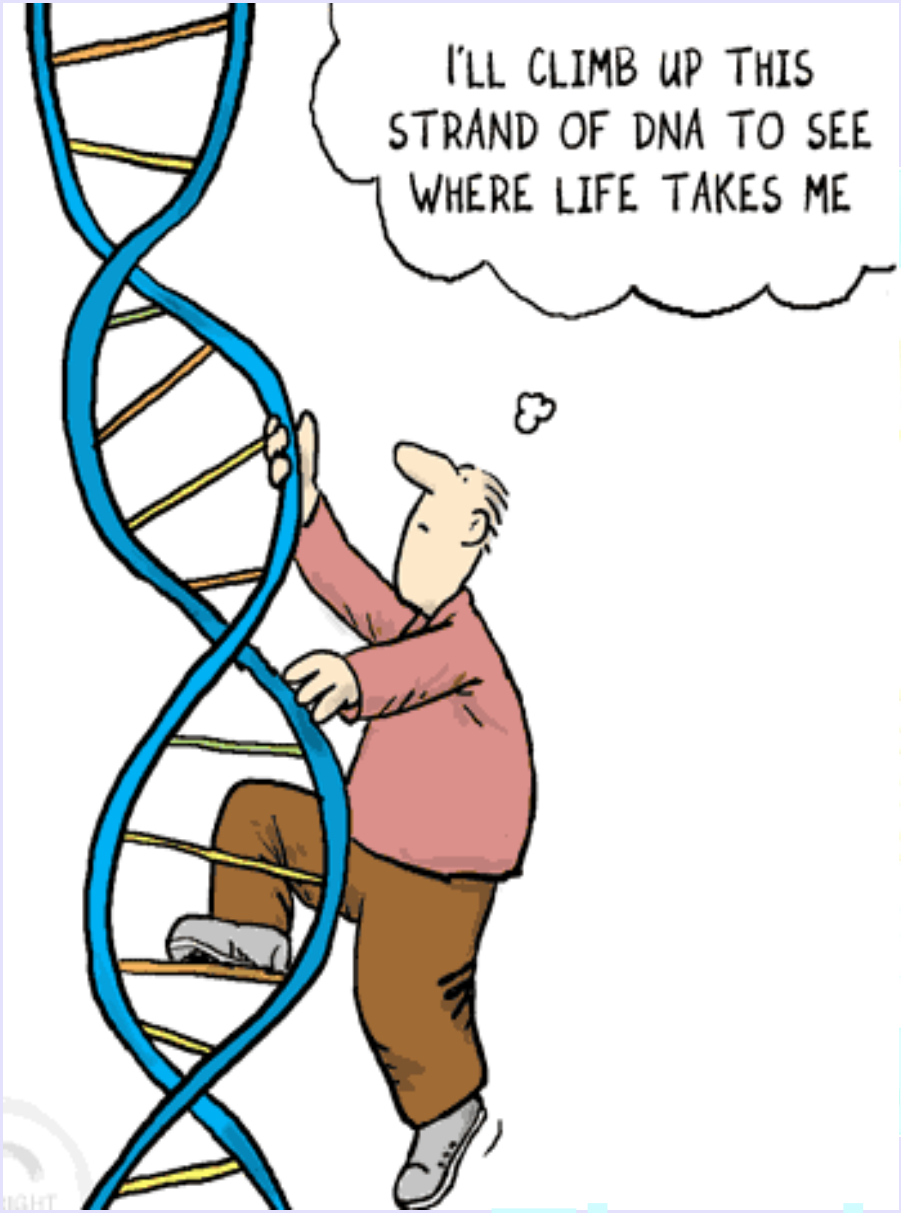
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Thank you